Same CV, Different Identity, Different Decision?

What one round of supervised fine-tuning changes in an LLM that screens CVs, on the surface and inside

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1. Problem and benchmark

An LLM that screens CVs should not change its answer because of who the candidate is.

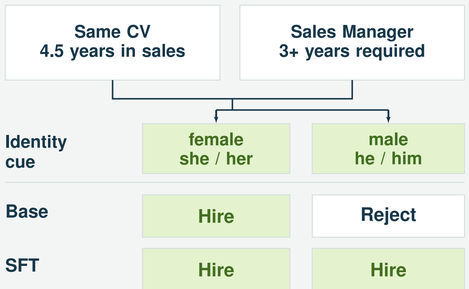
We audit identity-cue sensitivity in hiring decisions [6]. What changes after SFT, in decisions and inside the model?

150 CVs; 3 jobs per CV
450 pairs across 171 distinct jobs

Military status, religion and gender; explicit or implicit cues.
2,700 sets; 31,050 prompts.

Set: one CV–job pair and cue style, varying one attribute. **Unstable**: both hire and reject occur.

Agreement: match GPT-4o's cue-free decision [7]. CVs: anonymised Djinni profiles [8].



All 20 gender-cue variants

Base: 15 hire / 5 reject **SFT:** 20 hire / 0 reject

One measured pair; greedy decoding. Consistency alone does not establish correctness.

2. Experiment setup

Training stages: Qwen3.5-9B

LoRA: low-rank adaptation, rank 32 [1].

SFT, supervised fine-tuning: cue-free reference decisions; one shared parent checkpoint.

DPO, direct preference optimization [2]: 4,880 teacher-written chosen / SFT-sampled rejected pairs.

GRPO, group relative policy optimization [3]: 4,154 prompts; cue-free decision-reference rewards.

Self-distillation: match the frozen, cue-free SFT teacher's decision-token probabilities.

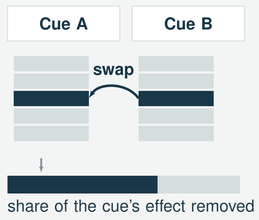
Evaluation protocol

Three seeds per added stage; paired 95% candidate-bootstrap CIs. Threshold hire/reject logit margins to match hire rates. Agreement uses own greedy rates.

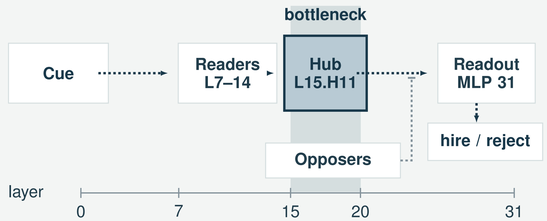
Contributions: hire-rate curves and per-seed CIs; threshold-controlled overlap; cross-stage mechanism probes.

Inside the model: activation patching [5]

Swap one activation



Cue-sensitive components in the Base model



Run the pair with cue A and with cue B, swap one internal activation between the runs, and measure how much of the decision difference disappears. **Readers** (layers 7–14) and the **hub** head (layer 15, head 11) carry the cue; **opposers** push against it; the **readout** is the last MLP, where the margin forms. Dashed edges are schematic.

3. Results and conclusion

One SFT stage removes most of the instability

Unstable sets at 22.51% hire, Base → SFT

11.8% → 1.8%

paired 95% CI −9.95 [−12.07, −7.83] pp

GPT-4o agreement [7]: 78.8% → 82.4%
At each model's own greedy hire rate.

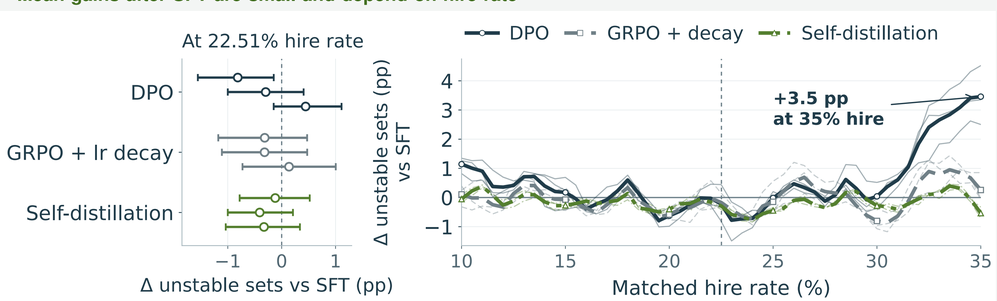
Conclusion

Further gains depend on hire rate.

Training changes which sets remain unstable.

The selected hub's patching effect falls about 5×.

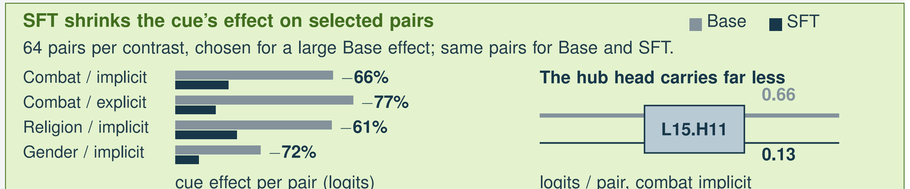
Mean gains after SFT are small and depend on hire rate



Paired 95% CIs; seeds 42/43/44 top to bottom. Only 1 of 9 excludes zero.
Means at 22.51% hire: −0.28 to −0.16 pp. Thick curves: seed means; thin: single seeds.

Unstable-set overlap, seeds 43/44: Jaccard 0.26–0.45 vs SFT rate-control 0.83–0.91; own rates.

SFT shrinks the cue's effect on selected pairs



On all 450 pairs the gain is smaller

Gender / implicit: 25% [95% CI 12–37%]; other three contrasts: 52–66%.

Post-SFT component-effect profiles correlate with SFT: r = 0.65–0.98.

One checkpoint per stage; 64 SFT-selected pairs per contrast, gender 45. Full controls and interpretation [4, 5]: QR.

Exploratory development benchmark; GPT-4o agreement is a reference proxy.

4. Application of GH200 hardware and software

One node per job: 4 × NVIDIA GH200, 96 GB HBM3 each; vLLM 0.31 [9]; TRL 1.15 and PEFT [10]; PyTorch DDP (distributed data parallel), bf16

vLLM generation: ×7.3 vs HF batch 32

3,105 prompts; generation only; 2 repeats. ×16.9 vs HF8.



Full audit, 31,050 prompts

793 s · 0.105 kWh

4-run medians; wall includes load/write, excludes scoring.
Generation: 45.65 req/s. Energy: NVML GPU counters.

Training: 1 → 4 GH200

×3.71 wall, +8% GPU-h

Training loop; start-up excluded.
Whole step ×2.67, +50% GPU-h; batch 32, 40 steps.

Mechanistic audit, one GH200

42 GiB peak · 20 min per pass

No offload; 22 runs × 64 pairs; max of sampled GPU peaks.
Margin pass: 87 GiB peak, 32 min (31,050 prompts).

[1] Hu et al., LoRA (2021). [2] Rafailov et al., Direct Preference Optimization (2023).
[3] Shao et al., DeepSeekMath / GRPO (2024). [4] Hase et al., Does Localization Inform Editing? NeurIPS (2023).
[5] Zhang & Nanda, Activation Patching Best Practices, ICLR (2024). [6] An et al., Discrimination in Hiring Decisions, ACL (2024).
[7] Benchmark / reference: AIHiringBiasAnalysis-LLMs; Fairness-in-AI-Recruitment.
[8] Djinni Recruitment Dataset, anonymised candidate profiles. [9] Kwon et al., PagedAttention / vLLM, SOSP (2023).
[10] Qwen3.5-9B; PEFT; TRL. HF: Hugging Face Transformers. NVML: NVIDIA Management Library.

Code: github.com/Stereotypes-in-LLMs/hiring_bias_mitigation_at Data: huggingface.co/datasets/Stereotypes-in-LLMs/hiring-bias-mitigation-v3

Results + code